



Preliminary Knowledge Eigenvalues and Eigenvectors

如果 v 是一个非零向量, 且 λ 是一个数

- If v is a nonzero vector and λ is a number such that

$$Av = \lambda v$$

then v is said to be an **eigenvector** of A with

eigenvalue λ . v 被称为矩阵 A 的特征向量, λ 被称为对应的特征值

Example

$$\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = 3 \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$\uparrow$ A $\uparrow$ V (eigenvectors) $\uparrow$ λ (eigenvalues)

当矩阵 A 乘特征向量 v 时, 结果向量 Av 只是 v 的一个缩放 (伸长或缩短), 方向保持不变。除非 $\lambda < 0$, 缩放因子就是特征值 λ 。

Preliminary Knowledge Eigenvalues and Eigenvectors

q Eigenvectors (for a square $m \times m$ matrix S)

对于一个 $m \times m$ 的方阵 S , 特征值 λ 的
求解基于以下方程:

$$Sv = \lambda v$$

$v \in \mathbb{R}^m \neq 0$ eigenvector eigenvalue $\lambda \in \mathbb{R}$

Example

$$\begin{pmatrix} 6 & -2 \\ 4 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 2 \\ 4 \end{pmatrix} = 2 \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

要使这个方程有非0解 v ,

q How many eigenvalues are there at most?

$$Sv = \lambda v \iff (S - \lambda I)v = 0$$

only has a non-zero solution if $|S - \lambda I| = 0$

with **at most m distinct λ values**

对于一个 $m \times m$ 矩阵, 通过求解这个 m 阶多项式, 最多可以得到 m 个不同的特征值

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PCA中通常处理“协方差矩阵”

Preliminary Knowledge Eigenvalues and Eigenvectors

正交性(orthogonal): 对于对称矩阵 S , 对于不同特征值 ($\lambda_1 \neq \lambda_2$) 的特征向量是正交的。
For symmetric matrices, eigenvectors for distinct eigenvalues are **orthogonal** 前提是对称矩阵 S ($S = S^T$)

$$Sv_{\{1,2\}} = \lambda_{\{1,2\}} v_{\{1,2\}}, \text{ and } \lambda_1 \neq \lambda_2 \quad v_1 \cdot v_2 = 0$$

All eigenvalues of a real symmetric matrix are **real**.

一个实数对称矩阵的所有特征值都是实数

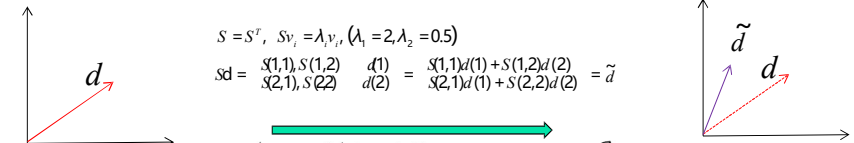
All eigenvalues of a **positive semidefinite** matrix are **non-negative**

一个半正定矩阵的所有特征值都是非负的

$$\forall w \in \mathbb{R}^n, w^T S w \geq 0, \text{ then if } Sv = \lambda v \quad \lambda \geq 0$$

positive semidefinite matrix: 对于任何向量 $w \in \mathbb{R}^n$, 都有 $w^T S w \geq 0$ 。若 S 是半正定, 则特征值 $\lambda \geq 0$

Preliminary Knowledge Eigenvalues and Eigenvectors

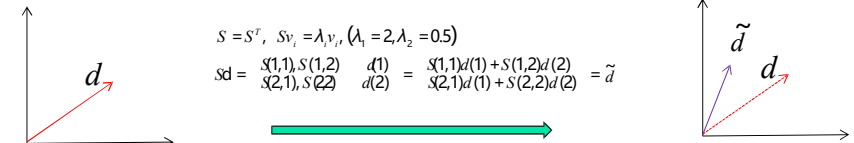


$S = S^T, S v_i = \lambda_i v_i, (\lambda_1 = 2, \lambda_2 = 0.5)$
 $S d = \begin{pmatrix} S(1,1) & S(1,2) \\ S(2,1) & S(2,2) \end{pmatrix} \begin{pmatrix} d(1) \\ d(2) \end{pmatrix} = \begin{pmatrix} S(1,1)d(1) + S(1,2)d(2) \\ S(2,1)d(1) + S(2,2)d(2) \end{pmatrix} = \tilde{d}$

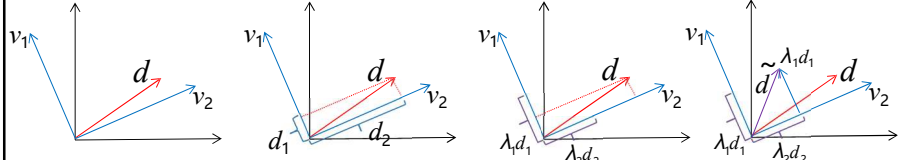
投影 (projection): 向量 d 可以分解为沿着特征向量 v_1 和 v_2 方向的分量 d_1 和 d_2 .
 缩放 (scaling): 当 S 位于 d 时, 其效果相当于将 d 投影到每个特征向量上, 然后用对应的特征值 λ_i 来缩放这些分量。
 在下图中, 分量 d_1 被 λ_1 缩放成 $\lambda_1 d_1$, 分量 d_2 被 λ_2 缩放成 $\lambda_2 d_2$.
 向量加法 (vector addition): 缩放后的分量相加, 就得到了变换后的向量 $\tilde{d}$.
 $S d = \tilde{d}$
 如果 v_1, v_2 是标准化的特征向量, 那么 $d = d_1 v_1 + d_2 v_2$, 且 $\tilde{d} = S d = S(d_1 v_1 + d_2 v_2) = d_1 (S v_1) + d_2 (S v_2) = d_1 (\lambda_1 v_1) + d_2 (\lambda_2 v_2)$.

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Preliminary Knowledge Eigenvalues and Eigenvectors



$S = S^T, S v_i = \lambda_i v_i, (\lambda_1 = 2, \lambda_2 = 0.5)$
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Projection on the eigenvectors Scaling with eigenvalues Vector addition

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Preliminary Knowledge Eigenvalues and Eigenvectors

Let $S = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ Real, symmetric.

Then $S - \lambda I = \begin{pmatrix} 2-\lambda & 1 \\ 1 & 2-\lambda \end{pmatrix}$ $(2-\lambda)^2 - 1 = 0$.

The eigenvalues are 1 and 3 (nonnegative, real). $\lambda = 1$ and 3 .

The eigenvectors are orthogonal (and real):

求特征向量: 将特征值 $\lambda=1$ 与 $\lambda=3$ 分别代入 $(S-\lambda I)V=0$

中求解对应的特征向量 $\begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}$

• result: $\begin{pmatrix} 1 \\ -1 \end{pmatrix}$ 和 $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$

这两个特征向量是正交的, 符合对称矩阵的性质.

Substitute these values
and solve for
eigenvectors.

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Diagonal Decomposition

Let $S \in \mathbb{R}^{m \times m}$ be a **square** matrix with **m linearly independent eigenvectors**

Theorem: Exists an **eigen decomposition**

$$S = U \Lambda U^{-1}$$

diagonal

Unique
for
distinct
eigen-
values

(matrix diagonalization theorem)

Columns of U are **eigenvectors** of S

Diagonal elements of Λ are **eigenvalues** of S

$$\Lambda = \text{diag}(\lambda_1, \dots, \lambda_m), \quad \lambda_i \geq \lambda_{i+1}$$

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Diagonal Decomposition

Write $\mathbf{U}$ with the eigenvectors as columns: $\mathbf{U} = \begin{bmatrix} \mathbf{v}_1 & \dots & \mathbf{v}_m \end{bmatrix}$

Then, $\mathbf{SU}$ can be written

$$\mathbf{SU} = \begin{bmatrix} S \mathbf{v}_1 & \dots & S \mathbf{v}_m \end{bmatrix} = \begin{bmatrix} \lambda_1 \mathbf{v}_1 & \dots & \lambda_m \mathbf{v}_m \end{bmatrix} = \begin{bmatrix} \mathbf{v}_1 & \dots & \mathbf{v}_m \end{bmatrix} \begin{bmatrix} \lambda_1 & & \\ & \dots & \\ & & \lambda_m \end{bmatrix}$$

Thus $\mathbf{SU} = \mathbf{U}\mathbf{\Lambda}$, or $\mathbf{U}^{-1}\mathbf{SU} = \mathbf{\Lambda}$

Therefore $\mathbf{S} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^{-1}$.

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Diagonal Decomposition

Recall $\mathbf{S} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$, $\lambda_1 = 1, \lambda_2 = 3$.

The eigenvectors $\begin{pmatrix} 1 \\ -1 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ form $\mathbf{U} = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$

Inverting $\mathbf{U}$, we have

$$\mathbf{U}^{-1} = \begin{bmatrix} 1/2 & -1/2 \\ 1/2 & 1/2 \end{bmatrix}$$

Recall
 $\mathbf{U}\mathbf{U}^{-1} = \mathbf{I}$.

$$\text{Then, } \mathbf{S} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^{-1} = \begin{bmatrix} 1 & 1 & 1 & 0 \\ -1 & 1 & 0 & 3 \end{bmatrix} \begin{bmatrix} 1/2 & -1/2 \\ 1/2 & 1/2 \end{bmatrix}$$

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Diagonal Decomposition

Let's divide $\mathbf{U}$ (and multiply $\mathbf{U}^{-1}$) by $\sqrt{2}$

$$\text{Then, } \mathbf{S} = \begin{matrix} & \begin{matrix} 1/\sqrt{2} & 1/\sqrt{2} & 1 & 0 & 1/\sqrt{2} & -1/\sqrt{2} \\ -1/\sqrt{2} & 1/\sqrt{2} & 0 & 3 & 1/\sqrt{2} & 1/\sqrt{2} \end{matrix} \\ \begin{matrix} \mathbf{Q} & \mathbf{\Lambda} & (\mathbf{Q}^{-1} = \mathbf{Q}^T) \end{matrix} \end{matrix}$$

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Symmetric Diagonal Decomposition

- **Theorem:** If $\mathbf{S} \in \mathbb{R}^{m \times m}$ is a **symmetric** matrix, there exists an **eigen decomposition**, where $\mathbf{Q}$ is **orthogonal**:

$$\mathbf{S} = \mathbf{Q} \mathbf{\Lambda} \mathbf{Q}^T$$

- $\mathbf{Q}^{-1} = \mathbf{Q}^T$
- Columns of $\mathbf{Q}$ are normalized eigenvectors
- Columns are orthogonal.
- (everything is real)

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Singular Value Decomposition

For an $m \times n$ matrix $\mathbf{A}$ of rank r , there exists a factorization (Singular Value Decomposition = **SVD**) as follows:

$$\mathbf{A} = \mathbf{U} \mathbf{\Sigma} \mathbf{V}^T$$

$\begin{matrix} & \nearrow & \uparrow & \nwarrow \\ \boxed{m \times m} & \boxed{m \times n, \text{nonnegative real}} & \boxed{n \times n} \end{matrix}$

The columns of $\mathbf{U}$ are orthogonal eigenvectors of $\mathbf{A}\mathbf{A}^T$.

The columns of $\mathbf{V}$ are orthogonal eigenvectors of $\mathbf{A}^T\mathbf{A}$.

Eigenvalues $\lambda_1 \dots \lambda_r$ of $\mathbf{A}\mathbf{A}^T$ are the eigenvalues of $\mathbf{A}^T\mathbf{A}$.

$$\sigma_i = \sqrt{\lambda_i}$$

$$\mathbf{\Sigma} = \text{diag}(\sigma_1 \dots \sigma_r) \leftarrow \text{Singular values.}$$

Singular Value Decomposition

- Illustration of SVD dimensions and

$$\begin{bmatrix} * & * & * \\ * & * & * \\ * & * & * \\ * & * & * \\ * & * & * \end{bmatrix} = \begin{bmatrix} * & * & * & * & * \\ * & * & * & * & * \\ * & * & * & * & * \\ * & * & * & * & * \\ * & * & * & * & * \end{bmatrix} \begin{bmatrix} \bullet & & & & \\ & \bullet & & & \\ & & \bullet & & \\ & & & & \\ & & & & \end{bmatrix} \begin{bmatrix} * & * & * \\ * & * & * \\ * & * & * \\ * & * & * \\ * & * & * \end{bmatrix}$$

$$\begin{bmatrix} * & * & * & * & * \\ * & * & * & * & * \\ * & * & * & * & * \\ * & * & * & * & * \\ * & * & * & * & * \end{bmatrix} = \begin{bmatrix} * & * & * \\ * & * & * \\ * & * & * \end{bmatrix} \begin{bmatrix} \bullet & & & & \\ & \bullet & & & \\ & & \bullet & & \\ & & & & \\ & & & & \end{bmatrix} \begin{bmatrix} * & * & * & * & * \\ * & * & * & * & * \\ * & * & * & * & * \\ * & * & * & * & * \\ * & * & * & * & * \\ * & * & * & * & * \end{bmatrix}$$

Singular Value Decomposition

$$\text{Let } A = \begin{pmatrix} 1 & -1 \\ 0 & 1 \\ 1 & 0 \end{pmatrix}$$

Thus $m=3, n=2$. Its SVD is

$$\begin{pmatrix} 0 & 2/\sqrt{6} & 1/\sqrt{3} \\ 1/\sqrt{2} & -1/\sqrt{6} & 1/\sqrt{3} \\ 1/\sqrt{2} & 1/\sqrt{6} & -1/\sqrt{3} \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{3} \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{pmatrix}$$

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Dimensionality Reduction

- One approach to deal with high dimensional data is by reducing their dimensionality.
- Project high dimensional data onto a lower dimensional **subspace** using linear or non-linear transformations.

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_N \end{pmatrix} \xrightarrow{\text{Reduce dimensionality}} \mathbf{y} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_k \end{pmatrix} \quad (K \ll N)$$

- Linear transformations are simple to compute

$$\mathbf{Y} = \mathbf{U} \mathbf{X} \quad (b_i = u_i^t a_i)$$

$\begin{matrix} \nearrow & \nearrow & \nearrow \\ k \times 1 & k \times d & d \times 1 \end{matrix} \quad (k \ll d)$

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Dimensionality Reduction

- **Find a basis in a low dimensional sub-space:**

- Approximate vectors by projecting them in a low dimensional sub-space:

(1) Original space representation:

$$x = a_1 v_1 + a_2 v_2 + \dots + a_N v_N$$

where $v_1, v_2, \dots, v_N$ is a basis in the original N-dimensional space

(2) Lower-dimensional sub-space representation:

$$\hat{x} = b_1 u_1 + b_2 u_2 + \dots + b_K u_K$$

where $u_1, u_2, \dots, u_K$ is a basis in the K-dimensional sub-space ($K < N$)

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Dimensionality Reduction

- If $K=N$, then $\hat{x} = x$
- Example ($K=N$):

$$v_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, v_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, v_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \quad (\text{standard basis})$$

$$x_v = \begin{bmatrix} 3 \\ 3 \\ 3 \end{bmatrix} = 3v_1 + 3v_2 + 3v_3$$

$$u_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, u_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, u_3 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \quad (\text{some other basis})$$

$$x_u = \begin{bmatrix} 3 \\ 3 \\ 3 \end{bmatrix} = 0u_1 + 0u_2 + 3u_3$$

thus, $x_v = x_u$

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Principal Component Analysis (PCA)

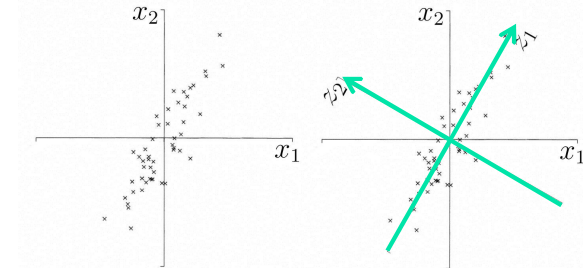
- Each dimensionality reduction technique finds an *appropriate transformation* by satisfying certain criteria (e.g., *information loss*, *data discrimination*, etc.)
- The goal of PCA is to reduce the dimensionality of the data while *retaining as much as possible of the variation present in the dataset*.

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Principal Component Analysis (PCA)

Motivation

- Find bases which has high variance in data
- Encode data with small number of bases with low MSE



- First PC is direction of maximum variance
- Subsequent PCs are orthogonal to 1st PC and describe maximum residual variance

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Principal Component Analysis (PCA)

Assume that $E[\mathbf{x}] = \mathbf{0}$ $a = \mathbf{x}^T \mathbf{q} = \mathbf{q}^T \mathbf{x}$ $\|\mathbf{q}\| = (\mathbf{q}^T \mathbf{q})^{1/2} = 1$

$$\begin{aligned} \Rightarrow \sigma^2 &= E[a^2] - E[a]^2 = E[a^2] \\ &= E[(\mathbf{q}^T \mathbf{x})(\mathbf{x}^T \mathbf{q})] = \mathbf{q}^T E[\mathbf{x} \mathbf{x}^T] \mathbf{q} = \mathbf{q}^T \mathbf{R} \mathbf{q} \end{aligned}$$

Find $\mathbf{q}$'s maximizing variance!!

It can be shown that variance is maximized when $\mathbf{q}$ is the principal component of $\mathbf{R}$.

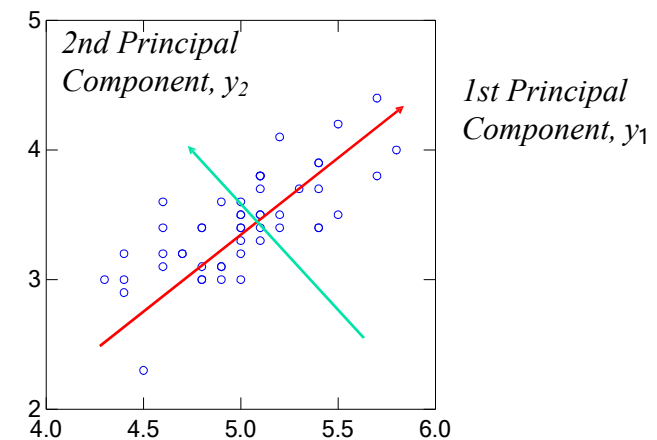
Principal component $\mathbf{q}$ can be obtained by **eigenvector decomposition**:

$$\mathbf{R} = \mathbf{Q} \mathbf{\Lambda} \mathbf{Q}^T, \quad \mathbf{Q} = [\mathbf{q}_1, \mathbf{q}_2, \dots, \mathbf{q}_j, \dots, \mathbf{q}_m], \quad \mathbf{\Lambda} = \text{diag}[\lambda_1, \lambda_2, \dots, \lambda_j, \dots, \lambda_m]$$

$$\mathbf{R} \mathbf{q}_j = \lambda_j \mathbf{q}_j \quad j = 1, 2, \dots, m \quad \Rightarrow \quad \mathbf{R} \mathbf{q} = \lambda \mathbf{q}$$

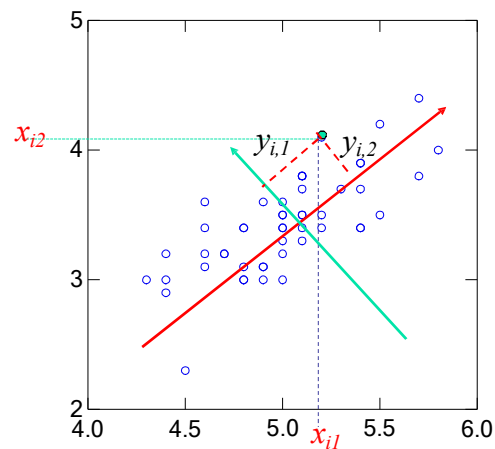
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Principal Component Analysis (PCA)



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Principal Component Analysis (PCA)



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Principal Component Analysis (PCA)

■ Advantage

- Reduce the dimension of the original data
 - reduce time consumption in the training process, and improve efficiency
- Discard some information of the original data
 - if this information is noise

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Principal Component Analysis (PCA)

■ Limitation

- Discard some information of the original data
 - if the discarded information is important, it is not appropriate to apply PCA
- The meaning of the principal component
 - PC or basis may not be interpretable
- Linear model of PCA
 - not suitable for nonlinear problem
- Assume first PC has higher importance

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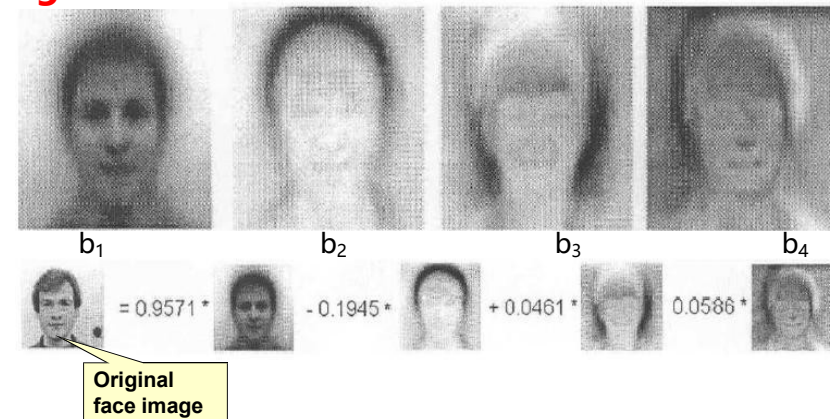
Case study: Eigenface

Face image: high-dimensional vector

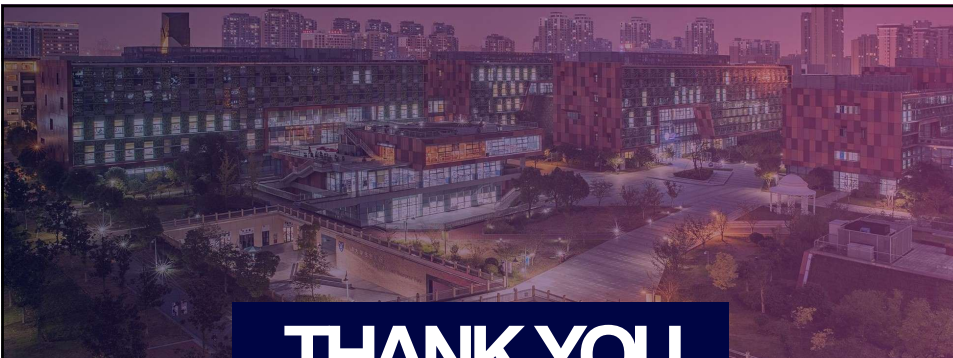
face image = linear combination of eigenvectors

The eigenvectors can be viewed as images.

Eigenfaces:



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